
NARAYANA
IIT ACADEMY
INDIA

40+ YEARS OF EXCELLENCE

Sec: SR.IIT_*CO-SC(MODEL-A,B&C) Date:09-01-24
Time: 3HRS Max. Marks: 300

Name of the Student: ____________________ H.T. NO:

09-01-24_SR.STAR CO-SUPER CHAINA(MODEL-A,B&C)_JEE-MAIN_GTM-11(N)_SYLLABUS

**PHYSICS:** TOTAL SYLLABUS

**CHEMISTRY:** TOTAL SYLLABUS

**MATHEMATICS:** TOTAL SYLLABUS

| SUBJECT | MISTAKES | | |
|---|---|---|---|
| | JEE SYLLABUS Q'S | JEE EXTRA SYLLABUS Q'S | TOTAL Q'S |
| MATHS | | | |
| PHYISCS | | | |
| CHEMISTRY | | | |

**PHYSICS** **MAX.MARKS: 100**

## SECTION – I
## (SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

**Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.**

1. A student determined young's modulus of elasticity using the formula $Y = \frac{MgL^3}{4bd^3\delta}$. The value of $g$ is taken to be $9.8\, m/s^2$, without any significant error, his observations are as following.

| Physical quantity | Least count of the equipment used for measurement | Observed value |
|---|---|---|
| Mass $(M)$ | 1 g | 2 kg |
| Length of bar $(L)$ | 1 mm | 1 m |
| Breadth of bar $(b)$ | 0.1 mm | 4 cm |
| Thickness of bar $(d)$ | 0.01 mm | 0.4 cm |
| Depression $(\delta)$ | 0.01 mm | 5 mm |

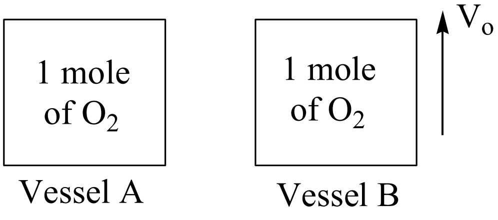
Then, the fractional error in the measurement of Y is

A) 0.0083 B) 0.0155 C) 0.155 D) 0.083

2. A box weighs 196 N on a spring balance at the north pole. Its weight recorded on the same balance if it is shifted to the equator is close to (Take $g = 10\, ms^{-2}$ at the north pole and the radius of the earth = 6400 km)

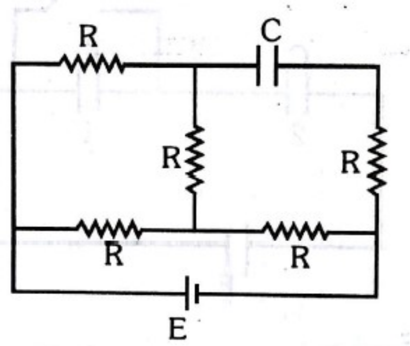
A) 195.66 N B) 194.66 N C) 194.32 N D) 195.32 N

3. Two identical vessels A and B contain one mole of $O_2$ each. The pressure in vessel A is '$P_0$' and that in B is 'P'. The r.m.s velocity of molecules in both the vessels are equal. The vessel A is at rest and B is moving with constant speed $V_0$, then which of the following is correct.



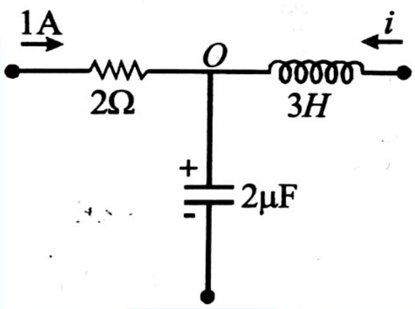

A) $P > P_0$ B) $P < P_0$ C) $P = P_0$ D) $P = P_0\left[\frac{V_{rms}^2 + V_0^2}{V_{rms}^2}\right]$

4. In the given circuit, the potential difference across the capacitor in steady state is 12V. Each resistance is of $3\Omega$. The cell is ideal. The emf of the cell as




A) 15V B) 9V C) 12V D) 24V

5. The potential difference (V) across the $2\mu F$ capacitor increases with time, and $\frac{dV}{dt} = 1V/s$ and $\frac{d^2V}{dt^2} = 2V/s^2$ at particular instant. The potential difference across the 3H inductor is___(Assume current through resistor is constant)




A) $4\mu V$ B) $12\mu V$ C) $6\mu V$ D) $8\mu V$

6. When voltage $v_s = 200\sqrt{2}\sin(\omega t + 15^0)$ is applied to an AC circuit the current in the circuit is found to be $i = 2\sin\left(\omega t + \frac{\pi}{4}\right)$ then average power consumed in the circuit is

A) 200 watt B) $400\sqrt{2}$ watt C) $100\sqrt{6}$ watt D) $200\sqrt{2}$ watt

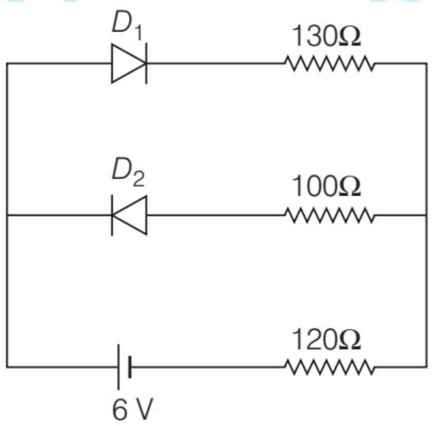
7. In the young's double-slit experiment, when a glass – plate (refractive index 1.5) of thickness $t$ is introduced in the path of one of the interfering beams (wavelength $\lambda$), the intensity at the position where the central maximum occurred previously remains unchanged. The minimum thickness of the glass-plate is

A) $2\lambda$ B) $\frac{2\lambda}{3}$ C) $\frac{\lambda}{3}$ D) $\lambda$

8. A sonometer wire resonates with a given tuning fork forming standing waves with five antinodes between the two bridges when a mass of 9kg is suspended from the wire. When this mass is replaced by mass M, the wire resonates with the same tuning fork forming three antinodes for the same positions of the bridges. The value of M is

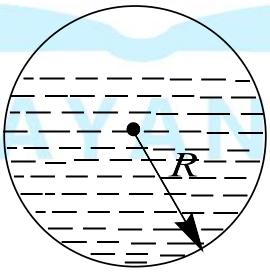
A) 25 kg B) 5 kg C) 12.5 kg D) 1/25 kg

9. The circuit contains two diodes each with a forward resistance of $50\Omega$ and with infinite reverse resistance. If the battery voltage is 6V, the current through the $120\Omega$ resistance is…..mA




A) 5 B) 10 C) 15 D) 20

10. A spherical glass vessel filled with liquid is kept in uniform gravity. Horizontal surface represents meniscus of liquid. Now complete system is taken to gravity free space. C is the center of sphere.




A) Vessel and liquid is wetting combination

B) Finally Liquid forms a drop

C) No effect on the liquid system

D) Finally liquid spreads over entire surface of vessel

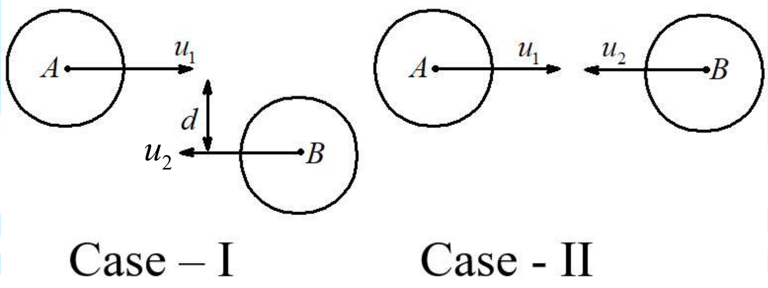
11. A plane electromagnetic wave of frequency 500 MHz is travelling in vacuum along y-direction. At a particular point in space and time, $B = 8.0\times10^{-8}\hat{z}T$ . The value of electric field at this point is (speed of light $= 3\times10^{8} ms^{-1}$ $\hat{x},\hat{y},\hat{z}$ are unit vectors along $x, y$ and z-direction)

A) $-24\hat{x}\dfrac{V}{m}$ B) $2.6\hat{x}\dfrac{V}{m}$ C) $24\hat{x}\dfrac{V}{m}$ D) $-2.6\hat{x}\dfrac{V}{m}$

12. A particle is taken from point A to point B under the influence of a force field. Now it is taken back from B to A and it is observed that the work done in taking the particle from A to B is not equal to the work done in taking it from B to A. If $W_{nc}$ and $W_c$ is the work done by non – conservative forces and conservative forces present in the system respectively, $\Delta U$ is the change in potential energy, $\Delta k$ is the change in kinetic energy, then choose the correct option

A) $W_C - \Delta U = \Delta K$ B) $W_C = +\Delta U$ C) $W_{nc} + W_c = \Delta K$ D) $W_{nc} - \Delta U = -\Delta K$

13. The following set of figures show two cases of collision between two balls. Let $\vec{u}_1$ and $\vec{u}_2$ be the velocities before collision and $\vec{v}_1, \vec{v}_2$ be the velocities after collision.




The coefficient of restitution, $e = \dfrac{|\vec{v}_2 - \vec{v}_1|}{|\vec{u}_1 - \vec{u}_2|}$

Student-A: Equation for 'e' holds for both cases as e is property of material of the colliding bodies.

Student-B: Equation for 'e' holds for case II only.

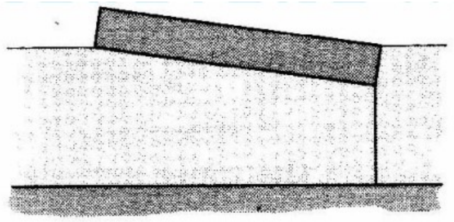
A) Student A is incorrect, student B is correct

B) Student A is correct, student B is incorrect

C) Both are incorrect

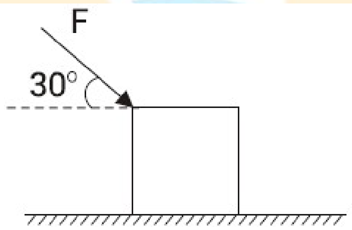
D) Both are correct

14. A uniform wooden plank floating on water is tied to the bottom of the pool such that in static equilibrium the diagonal plane of symmetry coincides with water surface, as shown in figure. The specific gravity of the wood is_______



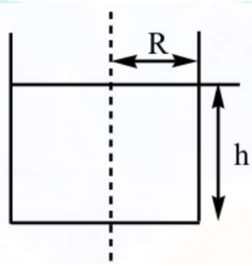
A) $\frac{1}{2}$ B) $\frac{1}{3}$ C) $\frac{2}{3}$ D) $\frac{3}{4}$

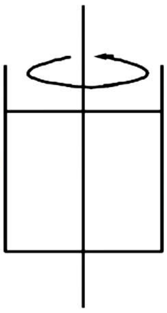
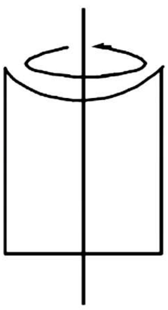
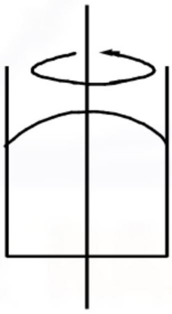
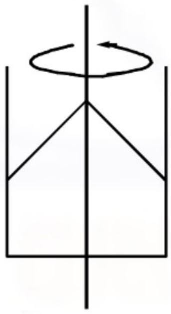
15. A body of mass 10 kg placed on a rough surface is pushed by force F making an angle of $30^0$ to the horizontal. If the angle of repose (between the block and the surface) is also $30^0$ , then the magnitude of minimum force F required to move the body is equal to ___ ($g = 10\ m/s^2$)




A) 100 N B) $50\sqrt{2}N$ C) $100\sqrt{2}N$ D) 50 N

16. A container is filled partially with a non-viscous fluid of density $'\rho'$ upto a height 'h'. Initially, the entire system is at rest. Now, the container starts rotating with a constant angular velocity $\omega_o$ about its vertical axis. Choose the appropriate plot from below which depicts the shape of free surface of the fluid after a long time.




A)  B)  C)  D) 

17. One mole of an ideal monoatomic gas is expanded till the temperature of gas is doubled under the process $TV^2 =$const (T is temperature and V is volume of gas). The initial temperature of gas is 400 K, the total work done in the process is

A) –400R B) –600R C) –300R D) –200R

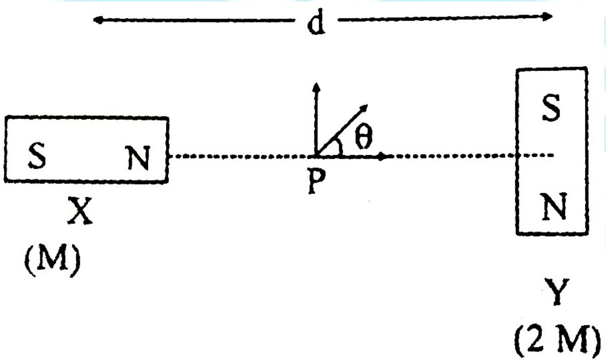
18. In a reactor, 2 kg of $_{92}U^{235}$ fuel is fully used up in 30 days. The energy released per fission is 200 MeV. Given that the Avogadro number, $N = 6.023\times10^{26}$ per kilo mole and $1eV = 1.6\times10^{-19}J$. The power output of the reactor is close to

A) 125 MW B) 35 MW C) 63 MW D) 54 MW

19. A closed organ pipe has a fundamental frequency of 1.5 kHz. The number of overtones that can be distinctly heard by a person with this organ pipe will be. (Assume that the highest frequency a person can hear is 20 kHz)

A) 7 B) 5 C) 6 D) 4

20. Two magnetic dipoles X and Y are placed at a separation d, with their axes perpendicular to each other. The dipole moment of Y is twice that of X. A particle of charge q is passing through their midpoint P, at angle $\theta = 45^0$ with the horizontal line, as shown in figure. What would be the magnitude of force on the particle at that instant? (d is much larger than the dimensions of the dipole)




A) $\left(\frac{\mu_0}{4\pi}\right)\frac{M}{\left(\frac{d}{2}\right)^3}\times qv$ B) 0 C) $\sqrt{2}\left(\frac{\mu_0}{4\pi}\right)\frac{M}{\left(\frac{d}{2}\right)}\times qv$ D) $\left(\frac{\mu_0}{4\pi}\right)\frac{2M}{\left(\frac{d}{2}\right)^3}\times qv$

## SECTION-II
## (NUMERICAL VALUE ANSWER TYPE)

This section contains 10 questions. The answer to each question is a Numerical value. If the Answer in the decimals , **Mark nearest Integer only. Have to Answer any 5 only out of 10 questions** and question will be evaluated according to the following marking scheme:

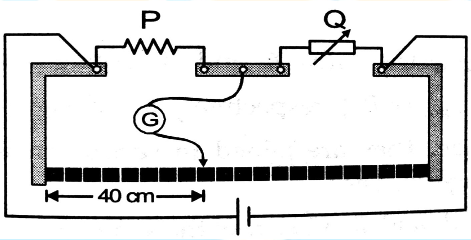
**Marking scheme: +4 for correct answer, -1 in all other cases.**

21. A stone is dropped from the top of a building. When it crosses a point 5m below the top, another stone starts to fall from a point 25 m below the top. Both stones reach the bottom of building simultaneously. The height of the building is (in meters) $\left(g=10\,m/s^2\right)$

22. A certain mass of a solid exists at its melting temperature of $20^0C$. When a heat Q is added $\frac{4}{5}$ of the material melts. When an additional Q amount of heat is added the material transforms to its liquid state at $50^0C$. Find the ratio of specific latent heat of fusion (in J/g) to the specific heat capacity of the liquid (in $J\,g^{-1}\,{}^0C^{-1}$) for the material

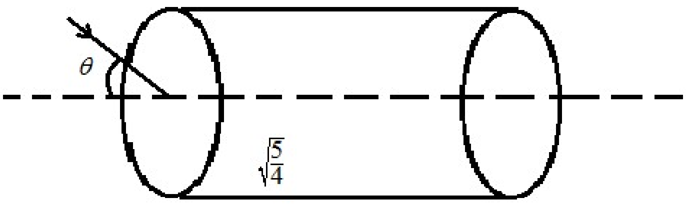
23. A positive point charge +q is placed at the origin. There is an electric field $E_{(x)}=E_0\left(2\frac{x}{d}+3\frac{x^2}{d^2}\right)$, that accelerates the point charge along the x-axis. The kinetic energy of the charge when it reaches the position x = 2d is $XqdE_0$ then find the value of $X$.

24. In a meter bridge, gaps are closed by two resistances P and Q and the balance point is obtained at 40cm. When Q is shunted by a resistance of $10\Omega$, the balance point shifts to 50 cm. Find the value of Q? in $\Omega$



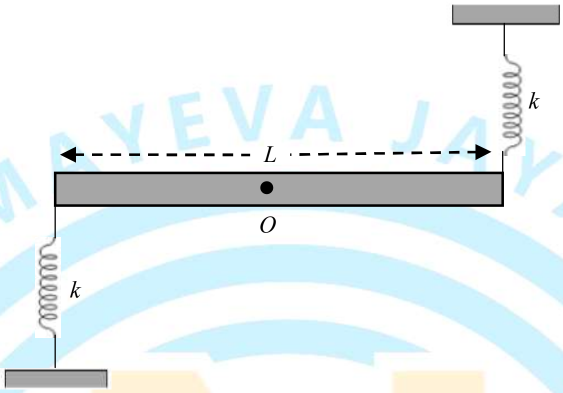

25. A particle of charge q and mass m starts moving from the origin under the action of an electric field $\vec{E}=E_0\hat{i}$ and magnetic field $\vec{B}=B_0\hat{i}$ with a velocity $\vec{v}=v_0\hat{j}$. The speed of the particle will become $2v_0$ after a time $t=\frac{\sqrt{P}mV_0}{qE_0}$ find the value P.

26. A monochromatic light ray is incident making an angle "$\theta$" with the axis of a transparent cylindrical fiber of refractive index $\sqrt{\frac{5}{4}}$ placed in vacuum as shown. Find the maximum value of $\theta$ (in degrees) so that the light entering the cylinder does not come out of the curved surface.



27. The total energy of an electron in the ground state of hydrogen atom is -13.6 eV. The magnitude potential energy (in eV) of an electron is the ground state of $Li^{2+}$ ion will be

28. A rod of mass m and length L is pivoted at its centre and can rotate in a vertical plane. Two springs each of force constant k are connected at its ends as shown in the figure. The time period of SHM of rod is $T = 2\pi\sqrt{\frac{m}{Pk}}$ for small oscillations. Find the value of P.




29. A rod of mass m and length $\ell$ rests on a smooth horizontal ground and is hinged at one of its ends. At the other end, a horizontal force F is applied whose magnitude is constant and the direction is always perpendicular the rod. When the rod rotates by $90^0$ angle, power supplied by this force at that instant is $\sqrt{\frac{xF^3\pi l}{m}}$. Find the value of $x$.


30. The energy flux of sunlight reaching the surface of the earth is $1.33\times10^3 W/m^2$. How many photons (nearly) per square metre are incident on the earth per second? Assume that the photons in the sunlight have an average wavelength of 550 nm (express the answer in multiples of $10^{21}$)

**CHEMISTRY** **MAX.MARKS: 100**

## SECTION – I
## (SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

**Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.**

31. Read the following Statements

1) First IP of Mg is less than that of Al

2) $O^{+}$ has lower electronegativity than that of $O^{-}$

3) Electron affinity of Cl is lesser than that of F

4) Second IP of oxygen is greater than that of N

Among the following, the correct option is

A) All are correct B) 1 only correct C) 1 and 2 only correct D) only 4 is correct

32. The correct statement regarding $ClO_n^-$ molecular ion is:

A) On decreasing value of 'n', $'Cl-O'$ bond order increases

B) On increasing value of 'n', $'Cl-O'$ bond length increases

C) On increasing value of 'n', oxidation number of central atom increase

D) On increasing value of 'n', hybrid orbitals on central atom increase

33. Which one of the following is a correct statement?

A) $Cr^{2+}$ is oxidising and $Mn^{3+}$ is reducing when both have $d^4$ configuration

B) Many copper(I) compounds are unstable in aqueous solution and undergo disproportionation as $2Cu^{+} \rightarrow Cu^{2+} + Cu$

C) Oxidation state of iron in $Fe_3O_4$ is fractional.

D) Actinoid contraction is lesser from element to element than lanthanoid contraction

34. Increasing value of magnetic moment of following species is

$I.\left[Fe(CN)_6\right]^{4-}$ $II.\left[Fe(CN)_6\right]^{3-}$ $III.\left[Cr(NH_3)_6\right]^{3+}$ $IV.\left[Ni(H_2O)_6\right]^{2+}$

A) $I < II < III < IV$ B) $IV < III < II < I$ C) $II < III < I < IV$ D) $I < II < IV < III$

35. The longest CO bond length will be with

A) $[Mn(CO)_6]^{+}$ B) $[V(CO)_6]^{-}$ C) $Cr(CO)_6$ D) $[Ti(CO)_6]^{2-}$

36. Assertion (A) : Boron always forms Covalent bond.

Reason (R) : The small size of $B^{3+}$ favours formation of covalent bond

1. Both A and R are true and R is the correct explanation of A
2. Both A and R are true but R is not the correct explanation of A
3. A is true but R is false
4. A is false but R is true

37. Match the Column – I with Column-II

| **COLUMN - I** | | **COLUMN – II** | |
|---|---|---|---|
| A | $NH_4Cl$ | P | Covalent bond |
| B | $HNC$ | Q | Ionic bond |
| C | Liquid $H_2O_2$ | R | Hydrogen bond |
| D | $CuSO_4.5H_2O$ | S | Co-ordinate bond |

A) $A \to p,q,s; B \to p,s; C \to p,r; D \to p,q,r,s$

B) $A \to p,r,s; B \to p,q; C \to q,r; D \to p,q,r,s$

C) $A \to p,r,s; B \to p,q; C \to q; D \to r,s$

D) $A \to p,r,s; B \to p,q,s; C \to q,s; D \to q,r,s$

38. Match **List – I** with **List – II** and select the correct answer using the code given below lists

| | **List – I** | | **List – II** |
|---|---|---|---|
| (P) | Eq.wt. $= \dfrac{\text{Molecular weight}}{33}$ | (1) | When $CrI_3$ oxidizes into $Cr_2O_7^{2-}$ and $IO_4^-$ |
| (Q | Eq.wt. $= \dfrac{\text{Molecular weight}}{27}$ | (2) | When $Fe(SCN)_2$ oxidizes into $Fe^{3+}$, $SO_4^{2-}$, $CO_3^{2-}$ and $NO_3^-$ |
| (R) | Eq.wt. $= \dfrac{\text{Molecular weight}}{28}$ | (3) | When $NH_4SCN$ oxidizes into $SO_4^{2-}$, $CO_3^{2-}$ and $NO_3^-$ |
| (S) | Eq.wt. $= \dfrac{\text{Molecular weight}}{24}$ | (4) | When $As_2S_3$ oxidizes into $AsO_3^-$ and $SO_4^{2-}$ |

A) $P \to 1; Q \to 2; R \to 4; S \to 3$
B) $P \to 2; Q \to 1; R \to 3; S \to 4$

C) $P \to 2; Q \to 1; R \to 4; S \to 3$
D) $P \to 1; Q \to 2; R \to 3; S \to 4$

39. Based on equation $E = -2.178\times10^{-18} J\left(\frac{Z^2}{n^2}\right)$, certain conclusions are written. Which of them is not correct?

A) The negative sign in equation simply means that the energy of electron bounded to the nucleus is lower than it would be if the electrons were at the infinite distance from the nucleus.

B) Larger the value of n, the larger is the orbit radius.

C) Equation can be used to calculate the change in energy when the electron changes orbit.

D) For n = 1, the electron has a more energy than it does for n = 6

40. For the reaction: $2A(g) + B(g) \longrightarrow 2D(g)$

$\Delta U^{o}_{298} = -2.5 kcal$ and $\Delta S^{o}_{298} = -10.5 cal/K$. Calculate approximate $\Delta G^{o}_{298}$ for the reaction, and predict whether the reaction may occur spontaneously.

A) 0.061 k cal, spontaneous B) -0.033 k cal, spontaneous

C) 0.061 k cal, non spontaneous D) 0.033 k cal, non-spontaneous

41. The equivalent conductivities of $K^+$, $Al^{+3}$ and $SO_4^{-2}$ ions x,y and Z $S\ cm^2 Eq^{-1}$ respectively. The $\Lambda^0 eq$ for $K_2SO_4\ Al_2(SO_4)_3.24H_2O$ (Potash Alum)

A) $\left(\frac{x}{4}+\frac{3y}{4}+z\right) Scm^2Eq^{-1}$ B) $x+3y+z\ Scm^2Eq^{-1}$

C) $\frac{x+y+z}{8}\ Scm^2Eq^{-1}$ D) $2x+3y+4z\ Scm^2Eq^{-1}$

42. Consider the following carbocations.

I. $C_6H_5\overset{+}{C}H_2$ II. $C_6H_5CH_2\overset{+}{C}H_2$ III. $C_6H_5\overset{+}{C}H\,CH_3$ IV. $C_6H_5\overset{+}{C}(CH_3)_2$

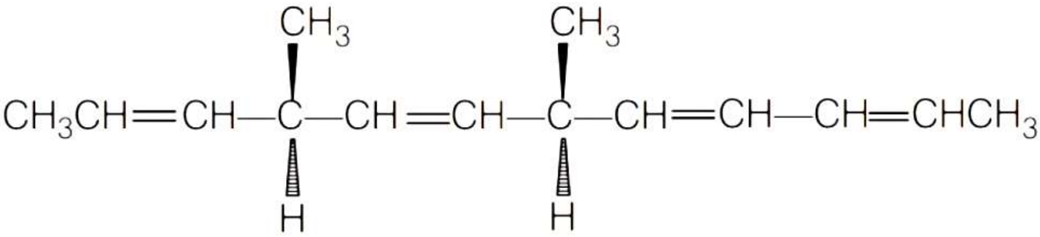
The correct sequence of the stability of these carbocation is

A) $II < I < III < IV$ B) $II < III < I < IV$ C) $III < I < II < IV$ D) $IV < III < I < II$

43. A mixture of ethyl iodide and n-propyl iodide is subjected to Wurtz reaction. The hydrocarbon which will not be formed is (exclude side reaction products)

A) Butane B) Propane C) Pentane D) Hexane

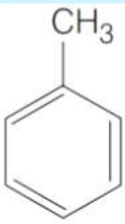
44. The number of optically active products obtained from the complete ozonolysis of the given compound is

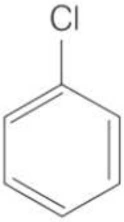


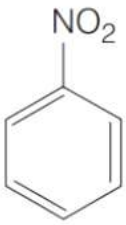
A) 0 B) 1 C) 2 D) 4

45.

I. 

II. 

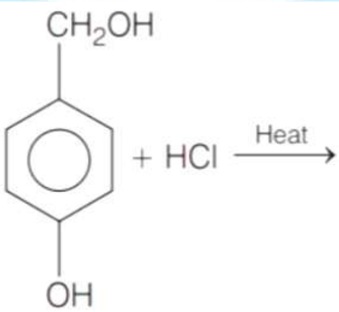
III. 

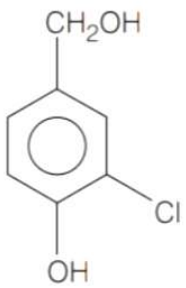
IV. 

From the above compounds correct order of reactivity in electrophilic aromatic substitution reactions will be

A) $II > I > III > IV$ B) $IV > III > II > I$ C) $I > II > III > IV$ D) $II > III > I > IV$

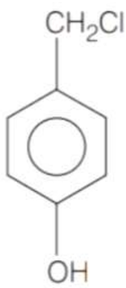
46. The major product in the following reaction.



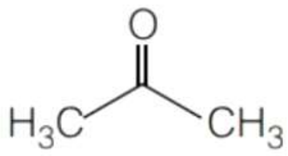
(a) 

(b) 

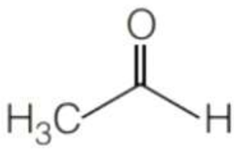
(c) 

(d) 

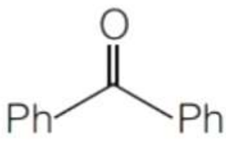
47. The order of reactivity of phenyl magnesium bromide with the following compounds is



(I)



(II)

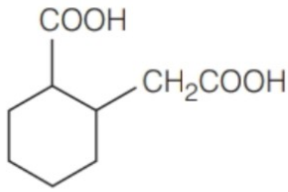


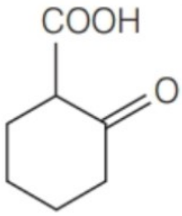
(III)

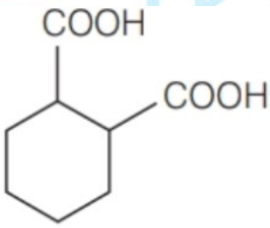
A) $II > III > I$ B) $I > III > II$

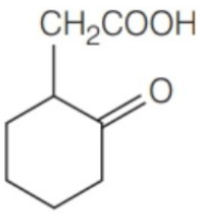
C) $II > I > III$ D) All react with the same rate

48. The compound that undergoes decarboxylation most readily just on heating is…

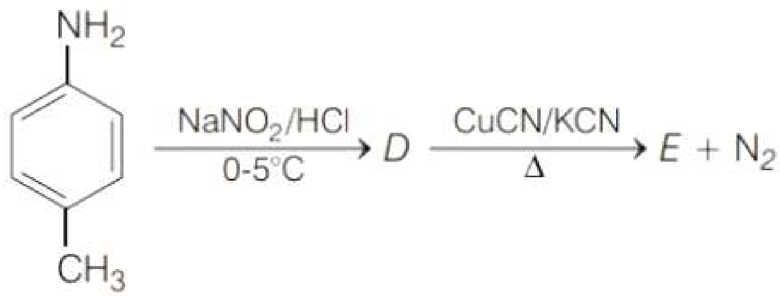
A) 


B) 

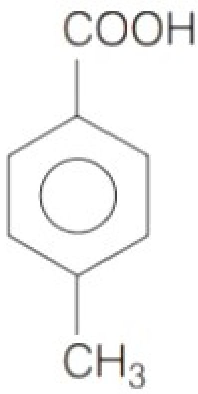

C) 

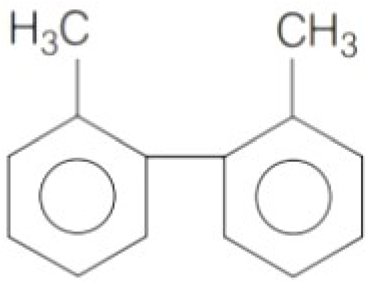

D) 


49. In the reaction,

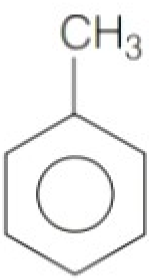



The product E is

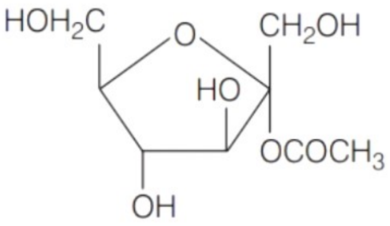
A) 

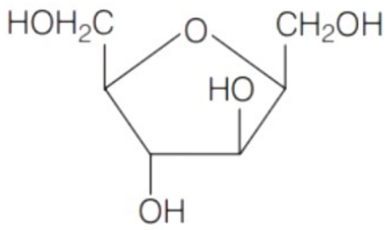

B) 

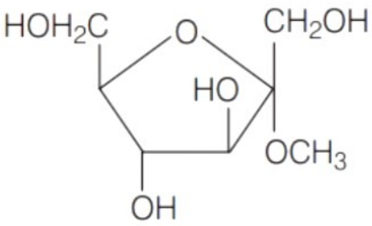

C) 

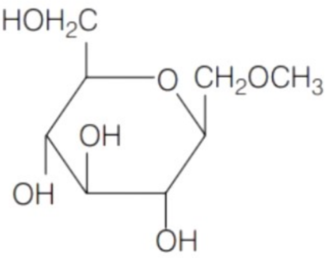

D) 

50. Which of the following compounds will behave as a reducing sugar in an aqueous KOH solution?

A) 

B) 

C) 

D) 

## SECTION-II
## (NUMERICAL VALUE ANSWER TYPE)

This section contains 10 questions. The answer to each question is a Numerical value. If the Answer in the decimals , **Mark nearest Integer only. Have to Answer any 5 only out of 10 questions** and question will be evaluated according to the following marking scheme:

**Marking scheme: +4 for correct answer, -1 in all other cases.**

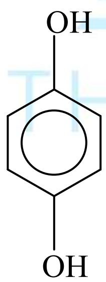
51. How many of the following reagents convert isopentyl alcohol into alkyl halide without any rearrangement?

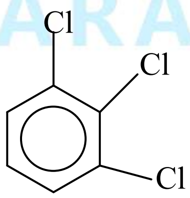
a) $HCl, ZnCl_2, \Delta$ b) $SOCl_2$ c) $PCl_5$ d) $PCl_3$ e) $PBr_3$ f) $HI$

52. How many of the following substances can act both as oxidizing and reducing agents

$H_3PO_2$, $H_3PO_3$, $H_3PO_4$, $HNO_2$, $SO_2$, $NO$, $N_2O_3$, $NO_2$, $SeO_2$, $TeO_2$

53. How many of the following molecules are non-polar?

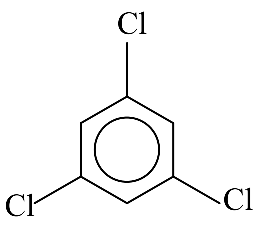
i) 

ii) 

iii) 

iv) 

v) $Cl-CH_2-Cl$

vi) 

vii) $HO-CH_2-CH_2-OH$

54. What is the molecular weight of the naturally occurring optically inactive $\alpha -$amino acid?

55. The number of revolutions made by an electron in one second in H – atom 2$^{nd}$ orbit is eight times of numbers of revolution made by electron in one second in n$^{th}$ orbit of H – atom, then n is ……….

56. Calculate the percentage dissociation of $H_2S_{(g)}$ If 0.1 mole of $H_2S$ is kept in 0.5 L vessel at 1000K. The value of $K_c$ for the reaction $2H_2S_{(g)} \rightleftharpoons 2H_{2(g)} + S_{2(g)}$ is $1.0\times10^{-7}$

57. A 4.0 molar aqueous solution of $NaCl$ is prepared and 500 mL of this solution is electrolysed. This leads to the evolution of chlorine gas at one of the electrodes. If the cathode is a Hg electrode, the maximum weight (g) of amalgam (Na – Hg) formed from this solution is : (atomic mass: Na=23, Hg=200)

58. Two substances A and B are present such that $[A]=4[B]$ and half-life of A is 5 minute and of B is 15 minute. If they start decaying at the same time following first order, how much time (minutes) later will the concentration of both of them would be same?

59. The total number of monochloro products obtained on chlorination of 3-Methylpentane in presence of sunlight is_____ (including stereo isomers).

60. How many of the following can give aldol condensation reaction?

$a) CH_3-CHO$ $b) CH_3-\underset{\underset{O}{\|}}{C}-CH_3$ $c) CH_3-CH_2-CHO$ $d) CH_3-COOH,$

$e) C_6H_5-CHO$ $f) CH_3-COOCH_3$ $g) C_6H_5-CH_2-CHO$ $h) (CH_3)_3C-CHO$

## MATHEMATICS MAX.MARKS: 100

### SECTION – I
### (SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

**Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.**

61. Let $I=\int_{\pi/4}^{\pi/3}\left(\frac{8\sin x-\sin 2x}{x}\right)dx$. Then

A) $\frac{2\pi}{3}<I<\frac{3\pi}{4}$ B) $\frac{\pi}{5}<I<\frac{5\pi}{12}$ C) $\frac{5\pi}{12}<I<\frac{2}{3}\pi$ D) $\frac{3\pi}{4}<I<\pi$

62. General solution of $x^2ydx=\left(x^3+y^3\right)dy$, is (where c being arbitrary constant)

A) $\frac{x^2}{2y^2}=ln|x|+c$ B) $\frac{x^2}{2y^2}=ln|y|+c$ C) $\frac{x^3}{3y^3}=ln|y|+c$ D) $\frac{x}{y}-2ln|x|+3ln|y|=c$

63. The set of values of k $(k\in R)$ such that det $(adj(adjA))$=16, where $A=\begin{bmatrix}k&1&2\\0&-1&1\\4&1&1\end{bmatrix}$, is equal to

A) $\{5,7\}$ B) $\{2,7\}$ C) $\{5,2\}$ D) $\{5,3\}$

64. Let $f(x)=3x^{10}-7x^8+5x^6-21x^3+3x^2-7$, then the value of $\lim_{h\to 0}\frac{f(1-h)-f(1)}{h^3+3h}$ is equal to

A) $\frac{-53}{3}$ B) $\frac{22}{3}$ C) $\frac{-22}{3}$ D) $\frac{53}{3}$

65. The complex number z which satisfies the equations $|z|=1$ *and* $\left|\frac{z-\sqrt{2}(1+i)}{z}\right|=1$ is

A) 1 B) 1+i C) $\frac{1+i}{\sqrt{2}}$ D) $\frac{-1-i}{\sqrt{2}}$

66. Let A and B be two sets each containing three elements then number of subsets of $A\times B$, each having at least two and at most 7 elements, is equal to

A) 502 B) 492 C) 456 D) 1002

67. If $\vec{V_1}=\hat{i}+\hat{j}+\hat{k}$ and $\vec{V_2}=a\hat{i}+b\hat{j}+c\hat{k}$ where $a,b,c\in\{-2,-1,0,1,2\}$, then number of possible non-zero vectors $\vec{V_2}$ such that $\vec{V_2}$ is perpendicular to $\vec{V_1}$ is

A) 10 B) 13 C) 15 D) 18

68. Let $y = y(x)$ be the solution curve of the differential equation $\sin(2x^2)\log_e(\tan x^2)dy + \left(4xy - 4\sqrt{2}x\sin\left(x^2 - \frac{\pi}{4}\right)\right)dx = 0, 0 < x < \sqrt{\frac{\pi}{2}}$, which passes through the point $\left(\sqrt{\frac{\pi}{6}}, 1\right)$. Then $\left|y\left(\sqrt{\frac{\pi}{3}}\right)\right|$ is equal to

A) 1 B) $\sqrt{3}$ C) 2 D) 3

69. If $f(x)$ is a differentiable function and satisfies $f(x+y) = f(x).\ f(y) \forall x, y \in R$ and $f(1) = 2$, then area enclosed by $3|x| + 2|y| \leq 8$ is (in sq.units)

A) $f(4)$ B) $\frac{1}{2}f(6)$ C) $\frac{1}{3}f(6)$ D) $\frac{1}{3}f(5)$

70. If roots of the equation $x^2 + ax + b = 0$ are 'c' and 'd' then one of the roots of the equation $x^2 + (2c+a)x + (c^2 + ac + b) = 0$ is always equal to $(c \neq d)$

A) c B) $d - c$ C) $2c$ D) $2d$

71. If $A_1, A_2, A_3$.........are in A.P then $\sum_{i=1}^{2n}(-1)^i\left(\frac{A_i + A_{i+1}}{A_i - A_{i+1}}\right)$ is equal to

A) 2n-1 B) n-1 C) -2n D) n+2

72. Let e be the eccentricity of hyperbola and f(e) be the eccentricity of its conjugate hyperbola then $\int\left(f(e) + f(f(e))\right)de = g(e)$ and $g(1) = \frac{1}{2}$, then $g(e) =$

A) $\frac{1}{2}\sqrt{e^2 - 1} + \frac{e^2}{2}$ B) $\frac{1}{2}\sqrt{e^2 + 1} + \frac{e^2}{2}$ C) $\sqrt{e^2 - 1} + \frac{e^2}{2}$ D) $\sqrt{e^2 + 1} + \frac{e^2}{2}$

73. Nine balls of the same size and colour, numbered $1, 2, \ldots..9$, were put into a packet. Now A draws a ball from packet, noted that it is of number a, and puts it back. Then B also draws a ball from the pocket and noted that it is of number $b$. Then probability for the inequality $a - 2b + 10 > 0$ to hold is

A) $\frac{52}{81}$ B) $\frac{59}{81}$ C) $\frac{60}{81}$ D) $\frac{61}{81}$

74. A scientist is weighing each of 30 fishes. Their mean weight worked out is 30 gm and a standard deviation of 2 gm. Later it was found that the measuring scale was misaligned and always under reported every fish weight by 2 gms (2gms less than the original weight of the each fish). The ratio of correct mean and standard deviation (in gm) of fishes are respectively

A) 16 B) 18 C) 22 D) 16.5

75. Let $z_1, z_2, z_3 \in C$ such that $|z_1| = 1, \left|z_2 - \left(1+\sqrt{3}\,i\right)\right| = 3$ and $\left|z_3 - 3 - 3\sqrt{3}\,i\right| = 7$. If $|3z_1 - 2z_2 - 4z_3|$ is maximum then the value of $|z_3 - z_1| + \frac{z_3 + 3z_1}{z_2}$ is equal to

A) 8 B) 12 C) 16 D) 18

76. The number of equivalence relations in set A = {a, b, c, d} is equal to

A) 3 B) 8 C) 12 D) 15

77. Each of three identical jewellery boxes has two drawers. In each drawer of the first box there is a gold watch. In each drawer of the second box there is a silver watch. In one drawer of the third box there is a gold watch while in the other there is a silver watch. The probability of selecting any drawer after selecting a box is half. If we select a box at random, open one of the drawers and find it to contain a silver watch, then the probability that the other drawer has the gold watch in it, is

A) $\frac{1}{3}$ B) $\frac{2}{3}$ C) $\frac{1}{2}$ D) $\frac{1}{4}$

78. Let ABC be a triangle with vertices at points $A(2,3,5), B(-1,3,2)$ and $C(\lambda,5,\mu)$ in three dimensional space. If the median through A is equally inclined with the axes, then $(\lambda,\mu)$ is equal to

A) $(7,5)$ B) $(10,7)$ C) $(5,7)$ D) $(7,10)$

79. If the function $f(x) = axe^{bx^2}$ have the local maximum value 1 at $x = 2$, then

A) $a = \frac{\sqrt{e}}{2}$ and $b = \frac{-1}{8}$ B) $a = \frac{-\sqrt{e}}{2}$ and $b = \frac{1}{8}$

C) $a = \frac{1}{8}$ and $b = \frac{\sqrt{e}}{2}$ D) $a = \frac{-\sqrt{e}}{2}$ and $b = \frac{-1}{8}$

80. Water is pouring into a conical vessel with vertex upwards and closed base at the rate of 3 cubic meters per minute through a tiny hole at the vertex. The radius of the base is 5m and the height is 10m. When the level is 7m from the base, the rate at which water level increases is (in m/min)

A) $\frac{18}{7\pi}$ B) $\frac{9\pi}{7}$ C) $\frac{4}{3\pi}$ D) $\frac{9}{7\pi}$

## SECTION-II
## (NUMERICAL VALUE ANSWER TYPE)

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81. The value of definite integral $\int_1^{\infty}\left(e^{x+1} + e^{3-x}\right)^{-1} dx$ is $\frac{\pi^a}{be^c}(a,b,c \in N)$ then the value of $a+b+c$ is

82. If $\lim\limits_{x\to\infty}\dfrac{\int_0^x\left(\tan^{-1}t\right)^2 dt}{\sqrt{1+x^2}}=\dfrac{\pi^2}{k}$, then k is

83. If $A=\int_1^{\sin\theta}\dfrac{t}{1+t^2}dt, B=\int_1^{\operatorname{cosec}\theta}\dfrac{1}{t\left(1+t^2\right)}dt, then \begin{vmatrix} A & A^2 & B \\ e^{A+B} & B^2 & -1 \\ 1 & A^2+B^2 & -1 \end{vmatrix}$ equals

84. The number of values of $x\in[0,n\pi], n\in I$ (the set of integers) that satisfy $\log_{|\sin x|}(1+\cos x)=2$ is______

85. For a biased die the probabilities for the different faces to turn up are given below

| **Face** | **1** | **2** | **3** | **4** | **5** | **6** |
|---|---|---|---|---|---|---|
| **Probability** | **0.1** | **0.32** | **0.21** | **0.15** | **0.05** | **K** |

This die is tossed and you are told that either face 1 or face 2 has turned up. If the probability that it is face 1, is $\frac{a}{b}$ where a and b $(a,b\in\mathrm{N})$ are coprime to each other then $a+b=$

86. If for positive integers $r>1, n>2$, the coefficients of the $(3r)^{th}$ and $(r+2)^{th}$ powers of $x$ in the expansion of $(1+x)^{2n}$ are equal, then $10\left(\dfrac{n_{r=5}}{n_{r=2}}\right)=....$ (where $n_{r=k}$ means value of $n$ when $r=k$ )

87. If the area bounded by the parabolas $y^2=4\alpha(x+\alpha)$ and $y^2=-4\alpha(x-\alpha)$, where $\alpha>0$ is 48 square units then $\alpha$ is equal to

88. The value of $\tan\left(\sum\limits_{r=1}^{\infty}\tan^{-1}\left(\dfrac{4}{4r^2+3}\right)\right)$ is equal to

89. Let the equations of two ellipses be $E_1:\dfrac{x^2}{3}+\dfrac{y^2}{2}=1$ and $E_2:\dfrac{x^2}{16}+\dfrac{y^2}{b^2}=1$. If the product of their eccentricities is $\dfrac{1}{2}$, then the product of all possible lengths of the minor axis for all possible positions of ellipse $E_2$ is

90. Let $f(x)$ be a differentiable function satisfying $(x-y)f(x+y)-(x+y)f(x-y)=4xy\left(x^2-y^2\right)$ for all $x,y\in R$. If $f(1)=1$, and the area of the region bounded by the curves $y=f(x)$ and $y=x^2$ is $\frac{a}{b}$ $(a,b\in\mathrm{N},\text{G.C.D of }(a,b)\text{ is }1)$, then the value of $\dfrac{b}{a}$ is equal to